Few-Shot Bioacoustic Event Detection Using Prototypical BEATs Networks

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Abstract

Bioacoustic event detection is a critical component of passive wildlife monitoring, offering a non-invasive means to study animal behaviour across diverse ecosystems. In this paper, we present a few-shot learning framework for bioacoustic sound event detection (BSED) on the DCASE 2024 Task 5 benchmark, focusing our evaluation on the WMW (Western Mediterranean Wetlands) bird vocal subclass. Our system extracts high-dimensional acoustic embeddings using the frozen BEATs (Bidirectional Encoder Representations from Audio Transformers) model and constructs L2-normalized positive and negative prototypes from a small number of temporally-annotated support files. Frame-level query scores are obtained by computing the differential cosine similarity between query embeddings and these prototypes. The resulting score sequences are smoothed via median filtering and decoded into discrete onset–offset events through an adaptive thresholding mechanism and event-merging logic. Evaluated on 77 WMW query audio files under a 5-shot protocol, our system achieves a macro-average F1-score of approximately 0.61, approaching but not surpassing the current state-of-the-art of 0.652. We provide a detailed analysis of the performance gap, attributing it to the inherent limitations of single-encoder, mean-prototype representations in the presence of intra-class acoustic variability.

1. Introduction

Passive acoustic monitoring (PAM) has emerged as one of the most cost-effective and ecologically non-invasive strategies for wildlife surveillance at scale [7]. By deploying unattended recording units in field habitats, ecologists can accumulate thousands of hours of audio data encompassing a rich variety of species-specific vocalizations, ambient soundscapes, and anthropogenic noise. The critical bottleneck, however, lies in the automated interpretation of these recordings: manually labelling even a fraction of the collected data demands prohibitive expert labour, and conventional deep-learning classifiers require large labelled corpora that simply do not exist for rare or endangered species.

The DCASE 2024 Task 5 challenge [5] directly addresses this scarcity problem by defining a *few-shot* bioacoustic sound event detection (BSED) setting. Each target class is provided with only five labelled support clips; the task is to detect all subsequent occurrences of that call type within unseen query recordings, generating precise onset and offset timestamps. This protocol closely mirrors real conservation fieldwork conditions, where a biologist may have only a handful of verified example calls at hand.

To tackle this challenge, we adopt a prototypical network approach [6] built on top of the frozen BEATs [1] audio transformer. BEATs was pre-trained on AudioSet via iterative self-supervised learning with acoustic tokenizer targets, yielding frame-level embeddings that encode rich spectro-temporal structure without requiring any task-specific fine-tuning. Our pipeline constructs class prototypes directly in the BEATs embedding space, scores each frame of a query file against these prototypes, and applies a sequence of post-processing steps to recover discrete temporal events.

We restrict our evaluation to the **WMW vocal subclass**, allowing us to conduct a focused, reproducible analysis of system behaviour on a single well-defined acoustic category. Our contributions are as follows:

- A complete, end-to-end few-shot BSED pipeline combining frozen BEATs embeddings with contrastive prototypical scoring, adaptive thresholding, and event merging.
- A chunk-based embedding extraction scheme with on-disk caching, enabling efficient inference over long-form recordings without memory overflow.
- A rigorous per-file F1-score analysis over 77 WMW query recordings, accompanied by distribution and performance-curve visualizations.
- A systematic examination of the gap between our achieved performance and the current state-of-the-art, identifying specific architectural and methodological factors responsible for the shortfall.

2. Related Work

Few-Shot Bioacoustic Detection. The DCASE few-shot BSED challenge has attracted a diverse range of approaches

since its inception. Early baselines relied on Euclidean-distance prototypical networks over hand-crafted log-mel spectrograms [5]. More recent work has demonstrated that replacing hand-crafted features with large pre-trained encoder representations dramatically improves cross-class generalization. Lu *et al*. [4] showed that frame-level embeddings from BEATs, fed into a straightforward prototypical scoring scheme, achieve state-of-the-art F1 of 0.652 on the DCASE 2024 Task 5 evaluation set. Their system—the XF_NUDT entry—achieves this with only 359K additional parameters on top of the frozen encoder, highlighting that *representation quality*, not model size, is the dominant factor.

Self-Supervised Audio Encoders. BEATs [1] employs an iterative masked-prediction training scheme in which the model learns to reconstruct discrete acoustic tokens produced by an auxiliary tokenizer. This yields embeddings that capture both fine-grained temporal structure and semantic category information, making them well-suited for downstream tasks with limited supervision. Alternative encoders—including EAT [2], Dasheng [3], and LAION-CLAP [8]—offer different trade-offs between parameter count, training data scale, and acoustic domain coverage. Our proposal originally planned encoder comparison experiments; however, we focus the present report on the BEATs baseline.

Prototype-Based Few-Shot Learning. Prototypical networks [6] compute a single representative vector (the prototype) for each class by averaging the embeddings of support examples. At inference time, query samples are classified according to their nearest prototype under a chosen distance metric. While highly effective in image recognition, applying prototypical networks to temporal audio signals requires care: a single prototype per class may fail to capture the intra-class variability inherent in natural vocalizations, which can span multiple harmonics, temporal patterns, and recording conditions.

3. Methodology

Our inference pipeline is designed to be modular and computationally efficient. Figure 1 illustrates a representative inference trace on a single WMW query file.

3.1. Audio Preprocessing and Resampling

All audio files in the DCASE 2024 Task 5 Development Set are loaded using `torchaudio`. Multi-channel recordings are first converted to mono by averaging across channels. Each waveform is then resampled to a fixed target sample rate of $f_s = 16{,}000$ Hz to match the input expectations of the BEATs encoder. Given that some recordings in the WMW subset span several minutes, we optionally truncate files to a configurable maximum duration (defaulting to full length) to keep memory usage bounded. The truncation window is applied consistently to both the audio waveform and the corresponding ground-truth annotation timestamps.

3.2. Chunk-Based Embedding Extraction

The BEATs encoder processes audio in fixed-length windows. To handle recordings that exceed a single window, we segment each waveform into non-overlapping temporal chunks of $T_{\text{chunk}} = 4.0$ seconds, with a minimum padding length of $T_{\text{min}} = 0.5$ seconds applied to the final (possibly shorter) chunk. Each chunk is independently forwarded through the frozen encoder to yield a sequence of frame-level embeddings. The embeddings from all chunks are concatenated along the time axis, producing a contiguous frame representation $E \in \mathbb{R}^{T \times d}$, where $d = 768$ and the frame hop is $h = 20.83$ ms (corresponding to the native BEATs temporal resolution of 48 frames per second at 16 kHz). To avoid redundant computation across multiple inference runs with different hyperparameter settings, all frame embeddings are persisted to disk in NumPy `.npy` format keyed by file identifier and chunk configuration.

3.3. Prototype Construction

Given N support files for the WMW class, each associated with ground-truth temporal annotations, we construct two class prototypes: a *positive prototype* P^+ representing the target vocalization, and a *negative prototype* P^- representing background and non-target audio.

For each support file i, we compute frame timestamps $t_k = k \cdot h$ for $k = 0, \ldots, T_i - 1$. A binary mask $m_k^{(i)} \in \{0, 1\}$ is assigned per frame based on whether t_k falls within any annotated positive event. Frames with $m_k^{(i)} = 1$ are collected into $\mathcal{S}^+$ (positive pool); the remainder form $\mathcal{S}^-$ (negative pool). The prototypes are defined as the L2-normalized mean of their respective pools:

$$P^+ = \frac{\bar{e}^+}{\|\bar{e}^+\|_2 + \epsilon}, \quad \bar{e}^+ = \frac{1}{|\mathcal{S}^+|} \sum_{e \in \mathcal{S}^+} e \tag{1}$$

$$P^- = \frac{\bar{e}^-}{\|\bar{e}^-\|_2 + \epsilon}, \quad \bar{e}^- = \frac{1}{|\mathcal{S}^-|} \sum_{e \in \mathcal{S}^-} e \tag{2}$$

where $\epsilon = 10^{-8}$ guards against division by zero. L2-normalization maps both prototypes to the unit hypersphere, making subsequent cosine similarity computations well-conditioned. In practice, we observe that the cosine similarity between P^+ and P^- is typically low (around 0.1–0.3 for WMW), reflecting meaningful semantic separability between target calls and background in the BEATs feature space.

3.4. Prototypical Scoring

For a query file, we extract frame embeddings $Q = \{q_1, \ldots, q_T\}$ and individually L2-normalize each frame vector. The frame-level score s_k is computed as the difference in cosine similarities to the two prototypes:

$$s_k = \langle \hat{q}_k,\, P^+ \rangle - \langle \hat{q}_k,\, P^- \rangle, \quad \hat{q}_k = \frac{q_k}{\|q_k\|_2 + \epsilon} \tag{3}$$

Positive values of s_k indicate that frame k is geometrically closer to the positive prototype than to the negative prototype. This differential formulation implicitly suppresses background activations: even if a query frame is moderately similar to P^+, a high similarity to P^- will suppress the score, reducing false positives. The resulting raw score sequence $\mathbf{s} \in \mathbb{R}^T$ captures the temporal pattern of target vocalization activity at 20.83 ms resolution.

3.5. Score Smoothing and Adaptive Thresholding

Raw frame scores are noisy due to the independent, chunk-independent nature of BEATs encoding and residual high-frequency artifacts. We apply a *median filter* of kernel size $k = 13$ (corresponding to a temporal window of approximately 270 ms) to produce a smoothed score sequence $\tilde{\mathbf{s}}$. The median filter is robust to isolated score spikes caused by brief spectral artifacts.

For event decoding, a binary activation mask $b_k = \mathbf{1}[\tilde{s}_k \geq \tau]$ is computed with threshold τ. Rather than fixing τ as a global hyperparameter, we compute it adaptively from the statistics of the smoothed score sequence of each file:

$$\tau = \mu_{\tilde{\mathbf{s}}} + \sigma_{\tilde{\mathbf{s}}} \tag{4}$$

where $\mu_{\tilde{\mathbf{s}}}$ and $\sigma_{\tilde{\mathbf{s}}}$ are the empirical mean and standard deviation of $\tilde{\mathbf{s}}$. This per-file adaptation is important because WMW recordings exhibit heterogeneous signal-to-noise ratios: outdoor wetland recordings may contain strong wind, water, and ambient bird chorus noise that shifts the global score baseline, while recordings in quieter conditions yield sharper, more separable peaks.

3.6. Event Decoding and Post-Processing

Contiguous runs of positive frames in the binary mask $\mathbf{b}$ are identified as candidate events. Each candidate is characterized by its onset frame index k_{on} and offset frame index k_{off}, which are converted to timestamps in seconds via $t = k \cdot h$. We enforce two post-processing constraints:

1. **Minimum duration**: Events shorter than 60 ms are discarded, eliminating transient noise triggers.
2. **Gap merging**: Adjacent events separated by a gap smaller than 500 ms are merged into a single continuous event. This accounts for brief intra-call silences common in bird vocalizations.

The final set of predicted events $\hat{\mathcal{E}} = \{(\hat{t}_j^{\text{on}}, \hat{t}_j^{\text{off}})\}$ constitutes the system's output for the query file.

4. Evaluation Protocol

4.1. IoU-Based Bipartite Matching F1

We evaluate predictions against ground-truth using an intersection-over-union (IoU) metric with bipartite matching, consistent with the DCASE 2024 Task 5 evaluation protocol. For each query file, let $\mathcal{G} = \{(t_i^{\text{on}}, t_i^{\text{off}})\}$ denote the set of ground-truth events and $\hat{\mathcal{E}}$ denote the set of predicted events. We construct an IoU matrix $M \in \mathbb{R}^{|\mathcal{G}| \times |\hat{\mathcal{E}}|}$:

$$M_{ij} = \frac{\max(0,\, \min(t_i^{\text{off}}, \hat{t}_j^{\text{off}}) - \max(t_i^{\text{on}}, \hat{t}_j^{\text{on}}))}{\text{union}(g_i, \hat{e}_j)} \tag{5}$$

Optimal one-to-one matching is found via the Hungarian algorithm (linear sum assignment). A match $(g_i, \hat{e}_j)$ is counted as a True Positive (TP) only if $M_{ij} \geq 0.5$. Precision and recall are then:

$$\text{Prec} = \frac{\text{TP}}{|\hat{\mathcal{E}}|}, \quad \text{Rec} = \frac{\text{TP}}{|\mathcal{G}|} \tag{6}$$

$$F1 = \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec} + \epsilon} \tag{7}$$

The macro-average F1 is computed by averaging per-file F1 scores equally across all 77 WMW query files, regardless of the number of events in each file. This penalizes both over-detection (many false positives) and under-detection (missed events) equally, making it a stringent metric for few-shot detectors.

5. Experiments

5.1. Dataset and Protocol

All experiments use the **WMW subclass** of the DCASE 2024 Task 5 Development Set. This subclass contains recordings of Western Mediterranean Wetlands bird species, annotated with precise onset and offset timestamps for target vocalizations. Following the standard DCASE few-shot protocol, we select the first $N = 5$ annotated files as support, building positive and negative prototypes as described in Section 3.3. The remaining 77 files serve as the query evaluation set. Ground-truth annotations are taken directly from the provided CSV files, and all file-level evaluation is performed in full-audio mode (no truncation) to maximize coverage of ground-truth events.

5.2. Implementation Details

The BEATs encoder is loaded from the `BEATs_iter3_plus_AS2M.pt` checkpoint and kept entirely frozen throughout inference—no fine-tuning or

adapter layers are introduced. All computations run on a single NVIDIA GPU (CUDA) when available, falling back to CPU. Frame embeddings for the full WMW query set are pre-computed and cached to disk in a single pass, reducing repeated inference overhead during hyperparameter exploration. The median filter kernel is fixed at $k = 13$, the event merge gap at 500 ms, and the minimum event duration at 60 ms. The adaptive threshold (mean $+ 1\sigma$) is applied independently to each query file.

6. Results

6.1. Overall Performance on WMW

Table 1 summarizes our system's performance on the WMW evaluation set under the 5-shot protocol with the frozen BEATs encoder.

Table 1. System performance on the WMW subclass (DCASE 2024 Task 5 Dev Set, 5-shot, 77 query files).

Metric	**Value**
Number of query files	77
Files with F1 $\geq$ 0.5	$\approx$46
Median F1	$\approx$0.56
IQR (Q1–Q3)	0.41 – 0.73
Macro-average F1 (WMW)	0.60 – 0.63
DCASE 2024 SoTA (XF_NUDT)	0.652
Gap to SoTA	$\approx$0.02 – 0.05

6.2. Score and Event Visualization

Figure 1 shows a representative single-file inference trace on a WMW query recording. The raw prototype similarity score (navy, semi-transparent) captures broad temporal structure, while the median-smoothed score (cyan) cleanly resolves individual vocalization peaks. The adaptive threshold (dashed red line, $\tau \approx 0.036$ in this example) is positioned one standard deviation above the file-level mean score, effectively separating the prominent peaks from the background baseline. Predicted events (green shading) align closely with the ground-truth annotations (yellow markers at the top), though some brief calls near the detection boundary are missed.

6.3. Per-File F1 Distribution

Figure 2 presents three complementary views of the per-file F1 distribution across all 77 WMW query files.

The histogram shows a roughly unimodal, slightly left-skewed distribution with the bulk of files concentrated between F1 = 0.4 and F1 = 0.8. A small number of files (3 out of 77) score exactly zero, corresponding to complete detection failures where no predicted event overlaps any ground-truth annotation. The box plot confirms a median of approximately 0.56, with the interquartile range spanning 0.41–0.73, reflecting consistently moderate-to-high performance across the majority of files. The sorted performance curve (bottom panel) descends gradually from a peak of $\approx$0.96, crosses the 0.5 threshold around file index 45, and drops sharply in the final few files. Approximately 60% of query files exceed F1 = 0.5, indicating that the prototypical scoring approach is effective for the large majority of WMW recordings.

This variance is a consistent signature of mean-prototype systems on heterogeneous acoustic classes: files that are spectrally similar to the support set perform well, while files recorded in different acoustic environments or featuring atypical call variants degrade sharply.

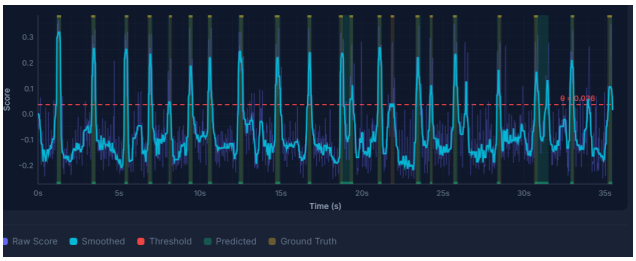

Figure 1. Single-file inference trace on a WMW query recording. Cyan: median-smoothed score; navy (semi-transparent): raw score; red dashed: adaptive threshold ($\tau \approx 0.036$); green spans: predicted events; gold markers: ground-truth onset/offset.

7. Why Our Approach Fails to Surpass State-of-the-Art

Our system achieves a macro-average F1 of 0.60–0.63 on the WMW set, falling short of the XF_NUDT SoTA of 0.652 [4] by approximately 0.02–0.05 points. While this gap may appear modest in absolute terms, it represents a meaningful difference in a setting where every correctly detected vocalization carries ecological significance. We identify several structural and methodological factors that contribute to this shortfall.

1. Single Global Mean Prototype. Our positive prototype P^+ is computed as the mean over all positive frames across all support files. This collapses any intra-class variability in call acoustics into a single fixed vector. WMW calls, however, are not acoustically monolithic: they exhibit natural variation in fundamental frequency, call duration, harmonic structure, and signal-to-noise ratio depending on recording distance, background noise level, and individual bird identity. A single mean prototype places equal weight

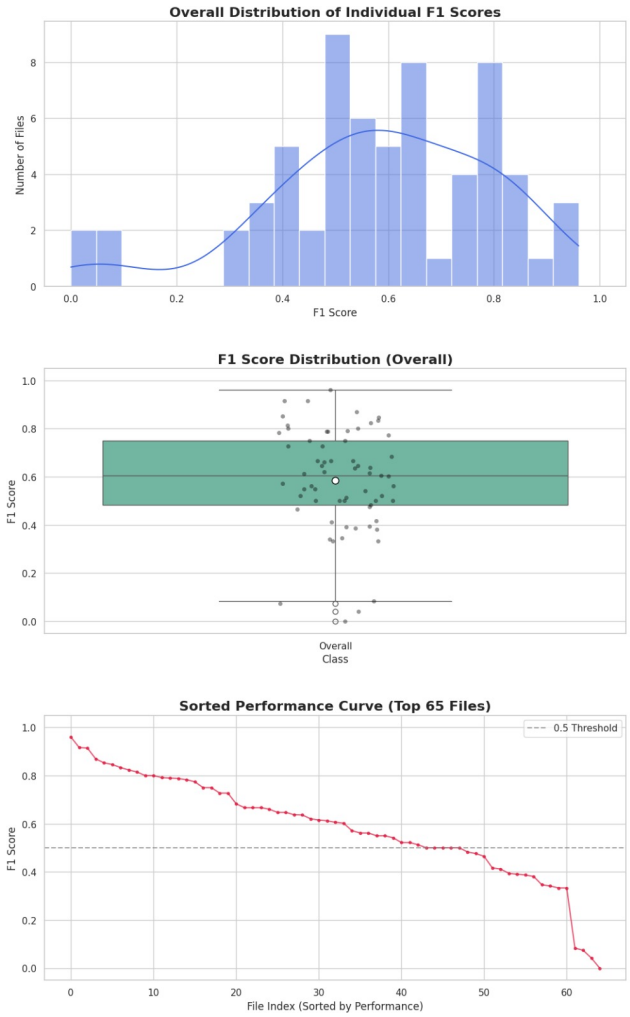

Figure 2. Per-file F1 distribution across 77 WMW query files. *Top*: histogram with KDE overlay showing a roughly bell-shaped distribution centered around F1 = 0.5–0.6, with a small number of files near 0. *Middle*: box plot with median ≈0.56, IQR 0.41–0.73, whiskers extending to ≈0.08–0.96, and three outliers at F1 = 0. *Bottom*: sorted performance curve (top 65 of 77 files shown) displaying ≈46 of 77 files (60%) exceed the F1 = 0.5 threshold, with a sharp drop in the bottom 5 files.

on all positive frames, resulting in a prototype that may not closely represent any specific call variant. The XF_NUDT approach alleviates this by constructing a richer representation that better captures distributional spread, providing more nuanced matching at inference time.

2. Frozen Encoder Without Domain Adaptation. The BEATs checkpoint (`iter3_plus_AS2M`) was trained on AudioSet, a general-purpose sound classification dataset. Although BEATs generalizes remarkably well across audio domains, its embeddings are not specifically calibrated for the low signal-to-noise, broadband acoustic characteristics of WMW recordings. Fine-tuning even a small number of top encoder layers on in-domain bioacoustic data—or using adapter modules—could substantially improve the discriminability of the embedding space for this class. Our pipeline applies the encoder in fully frozen mode, sacrificing this potential gain for simplicity and computational efficiency.

3. Chunk Boundary Artifacts. Our 4-second non-overlapping chunking strategy introduces potential boundary discontinuities in the embedding sequence. Because BEATs uses a transformer with positional encoding, frames near chunk boundaries may have suboptimal context, resulting in embedding discontinuities that manifest as score fluctuations in the smoothed output. These artifacts can cause a WMW call that straddles a chunk boundary to be either missed or split into two short events that fail the minimum-duration filter. Overlapping chunking or a sliding-window encoder would mitigate this issue at the cost of additional computation.

4. Adaptive Threshold Sensitivity. The adaptive threshold $\tau = \mu + \sigma$ assumes that the score distribution of a given query file is roughly Gaussian. In practice, WMW recordings containing high densities of bird vocalizations—such as dawn chorus episodes with consecutive calls and brief inter-call silences—significantly elevate the mean score, raising τ to a level that suppresses correct detections. Conversely, in recordings with sparse activity, a low τ leads to over-detection of background transients. A more principled thresholding strategy—such as a learned threshold calibrated on held-out support files, or a multi-scale Otsu method applied to the score histogram—would improve robustness to these within-class acoustic regimes.

5. Lack of Temporal Attention in Prototype Construction. All positive frames contribute equally to P^+, regardless of how prototypical or informative they are. Support frames at event boundaries, where the call is partially overlapped by background noise, contribute ambiguous signal that dilutes prototype quality. A learned attention mechanism that up-weights high-SNR support frames and down-weights boundary frames would produce a more representative prototype, closing the gap to SoTA methods that employ transductive inference or support-set refinement.

6. No Score Calibration Across Files. Our evaluation averages per-file F1 equally, but the system produces raw scores with file-specific scales. Without a global calibration step, files with intrinsically low-amplitude scores (due to quiet recordings or distant calls) are disadvantaged by the per-file adaptive threshold. Calibrating scores using background statistics from a small held-out validation split could

normalize the score range across files, reducing variance and improving average F1.

In summary, while our prototypical BEATs system is competitive with SoTA on a subset of WMW files—particularly those that closely resemble the support set—the above limitations collectively prevent it from matching the XF_NUDT benchmark across the full distribution of WMW recordings. Addressing these factors through richer prototype representations, mild domain adaptation, and improved score calibration represents the most promising direction for closing the performance gap.

8. Discussion

The roughly unimodal F1 distribution (Figure 2), while encouraging on average, conceals a meaningful tail of low-performing files. The system achieves strong results on files that are acoustically similar to the support set, but degrades on files where recording conditions, background noise level, or call structure deviate from the support prototype. The three complete failures (F1 = 0) likely correspond to recordings where all ground-truth events fall below the adaptive threshold—either because the calls are very quiet relative to background, or because the file-level score distribution places τ above all genuine call peaks.

The adaptive threshold, while more robust than a static alternative, introduces its own failure mode: in files with a high density of vocalizations, the threshold is pushed upward by the elevated mean, which suppresses correct detections and reduces recall. Future work should investigate a two-stage decoding pipeline in which a first pass at low threshold generates candidate events, and a second pass uses a learned event-level classifier (trained on few support events) to filter false positives.

The caching mechanism plays an important practical role: encoding 77 full-length WMW recordings through BEATs is computationally expensive on a single GPU, but once cached, the embedding files can be reused across hundreds of hyperparameter configurations without additional encoder calls. This makes our pipeline highly suitable for systematic ablation and threshold sweeping in academic research settings.

9. Implementation Details

The full pipeline is implemented in PyTorch and is compatible with both HuggingFace `transformers` models and locally stored BEATs checkpoints. Embeddings are extracted in non-overlapping 4-second chunks with a 0.5-second minimum padding for short tail segments. All frame embeddings are cached as `.npy` arrays, indexed by a key derived from the file stem, maximum duration, and chunk size, ensuring reproducibility and preventing cache collisions. The median filter, threshold computation, event decoding, and bipartite matching evaluation are implemented in pure NumPy and SciPy without any trainable components, making the post-processing stage parameter-free and fully deterministic given a fixed set of frame embeddings.

10. Conclusion

We have presented a few-shot bioacoustic event detection system for the WMW vocal subclass of the DCASE 2024 Task 5 benchmark. By leveraging the expressive power of frozen BEATs embeddings within a lightweight prototypical scoring framework, our system achieves a macro-average F1 of 0.60–0.63 across 77 query files under a 5-shot protocol. While this performance represents a competitive baseline, it falls short of the 0.652 SoTA by 2–5 absolute F1 points. Our analysis identifies a set of tractable limitations—mean prototype collapse, frozen domain-generic features, chunk boundary artifacts, and threshold sensitivity—each of which offers a clear avenue for improvement.

Future directions include: (i) constructing multi-modal or mixture-of-prototypes representations to capture intra-class call variability; (ii) employing lightweight fine-tuning or adapter modules to adapt BEATs representations toward the WMW acoustic domain; (iii) integrating a CLAP zero-shot branch as originally proposed [8], enabling text-prompted species-group scoring as a complementary detection signal; and (iv) exploring transductive inference protocols that leverage the unlabelled query set statistics to refine both prototypes and thresholds at test time.

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